

2.5

We define $(i\partial - m)\psi = \int_y i S_F^{-1}(x-y) \psi(y)$

$$\Rightarrow i S_F^{-1}(x-y) = (-i\partial_y - m) \delta^{(4)}(x-y)$$

The fermion propagator is the Greens function of the Dirac operator:

$$(i\partial_x - m) S_F(x-y) \equiv i \delta(x-y)$$

This means $\int_z S_F^{-1}(x-z) S_F(z-y) = \delta(x-y)$ must be true for S_F^{-1} to be inverse of S_F

Let's show this for $S_F(z-y) = \int \frac{d^4 q}{(2\pi)^4} \frac{i(q+m)}{q^2 - m^2 + i\epsilon} e^{-iq \cdot (z-y)}$

$$\begin{aligned} \int_z S_F^{-1}(x-z) S_F(z-y) &= -i \int_z (-i\partial_z - m) \delta^{(4)}(x-z) \int \frac{d^4 q}{(2\pi)^4} \frac{i(q+m)}{q^2 - m^2 + i\epsilon} e^{-iq \cdot (z-y)} \\ &= -i \int_z \int \frac{d^4 q}{(2\pi)^4} \frac{i(q+m)}{q^2 - m^2 + i\epsilon} (-i\partial_z - m) \delta^{(4)}(x-z) e^{-iq \cdot (z-y)} \\ &= -i \int_z \int \frac{d^4 q}{(2\pi)^4} \frac{i(q+m)}{q^2 - m^2 + i\epsilon} (i\partial_z - m) e^{-iq \cdot (z-y)} \delta^{(4)}(x-z) \\ &= \int \frac{d^4 q}{(2\pi)^4} \frac{(q+m)}{q^2 - m^2 + i\epsilon} (i\partial_x - m) e^{-iq \cdot (x-y)} \\ &= \int \frac{d^4 q}{(2\pi)^4} \frac{(q+m)}{q^2 - m^2 + i\epsilon} (i(-i)q - m) e^{-iq \cdot (x-y)} \\ &= \int \frac{d^4 q}{(2\pi)^4} \frac{(q+m)(q-m)}{q^2 - m^2 + i\epsilon} e^{-iq \cdot (x-y)} \end{aligned}$$

Now we integrate $\int_z$ Same integration by part as in the hints

$$\begin{aligned} \partial_x e^{-iq \cdot (x-y)} &= \partial_\mu e^{-iq \cdot (x-y)} \\ &= \partial_\mu (-iq_\mu) e^{-iq \cdot (x-y)} \\ &= -iq e^{-iq \cdot (x-y)} \end{aligned}$$

$$\begin{aligned} q \cdot q &= q^2 \\ \Rightarrow (q+m)(q-m) &= q^2 - m^2 \end{aligned}$$

The $i\epsilon$ prescription is there to calculate integral such that it doesn't diverge but here we have Numerator = Denominator without $i\epsilon$, so we don't need $i\epsilon \rightarrow \epsilon = 0$

$$\Rightarrow = \int \frac{d^4 q}{(2\pi)^4} \frac{q^2 - m^2}{q^2 - m^2} e^{-iq \cdot (x-y)} = \int \frac{d^4 q}{(2\pi)^4} e^{-iq \cdot (x-y)}$$

This integral is just a Dirac delta function

$$\Rightarrow = \delta^{(4)}(x-y) \Rightarrow \int_z S_F^{-1}(x-z) S_F(z-y) = \delta^{(4)}(x-y) \quad \square$$

we can drop (4) from $\delta^{(4)}(x-y)$ since x and y are 4-vectors

We have shown that $\int_z S_F^{-1}(x-z) S_F(z-y) = \delta(x-y)$

for $S_F(x-y) = \int \frac{d^4 q}{(2\pi)^4} \frac{i(\not{x}+m)}{q^2-m^2+i\epsilon} e^{-iq(x-y)}$ and $iS_F^{-1}(x-y) = (-i\not{\partial}_y - m) \delta^{(4)}(x-y)$

As such $S_F^{-1}(x-y)$ that is is defined as $iS_F^{-1}(x-y) = (-i\not{\partial}_y - m) \delta^{(4)}(x-y)$

is the inverse function of the fermion propagator $S_F(x-y) = \int \frac{d^4 q}{(2\pi)^4} \frac{i(\not{x}+m)}{q^2-m^2+i\epsilon} e^{-iq(x-y)}$

Exercise 2.6

Derive the generating functional for the free Dirac field,

$$Z_0[\bar{\eta}, \eta] = Z_0[0, 0] \exp \left[- \int d^4 x d^4 y \bar{\eta}(x) S_F(x-y) \eta(y) \right]$$

The generating functional of the free Dirac field is

$$Z_0[\bar{\eta}, \eta] = \int D\Psi D\bar{\Psi} \exp \left[i \int_x \bar{\Psi} (i\not{\partial} - m) \Psi + \bar{\eta} \Psi + \bar{\Psi} \eta \right]$$

We shift the fields by:

$$\underline{\Psi'(x) = -[\Psi(x) - i \int_y S_F(x-y) \eta(y)]}$$

This implies $\bar{\Psi}'(x) = -[\Psi'^{\dagger}(x) \gamma^0] = -\left[\bar{\Psi}(x) + i \left[\int_y \eta^{\dagger}(y) S_F^{\dagger}(x-y) \right] \gamma^0 \right]$

$$S_F(x-y) = i \int \frac{d^4 k}{(2\pi)^4} \frac{(\not{k}+m)}{k^2-m^2+i\epsilon} e^{-ik(x-y)} \Rightarrow S_F^{\dagger}(x-y) = -i \int \frac{d^4 k}{(2\pi)^4} \frac{(\not{k}(\gamma^{\mu})^{\dagger} + \gamma^0 m \gamma^0)}{k^2-m^2-i\epsilon} e^{-ik(y-x)}$$

we use $(\gamma^{\mu})^{\dagger} = \gamma^0 \gamma^{\mu} \gamma^0$

$$\Rightarrow S_F^{\dagger}(x-y) = -i \int \frac{d^4 k}{(2\pi)^4} \frac{\gamma^0 (\not{k}+m) \gamma^0}{k^2-m^2-i\epsilon} e^{-ik(y-x)} = -\gamma^0 S_F(y-x) \gamma^0$$

$$\Rightarrow \underline{\bar{\Psi}'(x) = -[\bar{\Psi}(x) - i \int_y \bar{\eta}(y) S_F(y-x)]}$$

$$\Rightarrow \bar{\eta} \Psi(x) \rightarrow -[\bar{\eta}(x) \Psi(x) - \int_y \bar{\eta}(x) S_F(x-y) \eta(y)]$$

$$\bar{\Psi}(x) \eta(x) \rightarrow -[\bar{\Psi}(x) \eta(x) - \int_y \bar{\eta}(y) S_F(y-x) \eta(x)]$$

Before we apply the shift we modify the functional. We use the relation

$$(i\not{\partial} - m) \Psi(x) = \int_y i S_F^{-1}(x-y) \Psi(y) \quad \text{to get}$$

$$Z_0[\bar{\eta}, \eta] = \int D\psi D\bar{\psi} \exp \left[i \int_x \int_y \bar{\psi}(x) i S_F^{-1}(x-y) \psi(y) + i \int_x \bar{\eta} \psi + \bar{\psi} \eta \right]$$

Let A shift of $\psi(x) \rightarrow \psi'(x)$ and $\bar{\psi}(x) \rightarrow \bar{\psi}'(x)$ is linear so

$$D\psi D\bar{\psi} \rightarrow D\psi' D\bar{\psi}' = -D\psi (-D\bar{\psi}) = D\psi D\bar{\psi}$$

So after the shift we have

$$Z_0[\bar{\eta}, \eta] = \int D\psi D\bar{\psi} \exp \left[i \int_x \int_y \bar{\psi}'(x) i S_F^{-1}(x-y) \psi'(y) + i \int_x \bar{\eta} \psi' + \bar{\psi}' \eta \right]$$

Let's evaluate the integrals in the exponential separately

$$\int_x \bar{\eta} \psi' = \int_x - \left[\bar{\eta}(x) \psi(x) - i \int_y \bar{\eta}(x) S_F(x-y) \eta(y) \right]$$

$$= - \int_x \bar{\eta} \psi + i \int_x \int_y \bar{\eta}(x) S_F(x-y) \eta(y)$$

$$\int_x \bar{\psi}' \eta = \int_x - \left[\bar{\psi}(x) \eta(x) - i \int_y \bar{\eta}(y) S_F(y-x) \eta(x) \right]$$

$$= - \int_x \bar{\psi} \eta + i \int_x \int_y \bar{\eta}(y) S_F(y-x) \eta(x)$$

$$= - \int_x \bar{\psi} \eta + i \int_x \int_y \bar{\eta}(x) S_F(x-y) \eta(y)$$

By renaming one of variables the integral on right becomes:

$$\int_y \int_x \bar{\eta}(x) S_F(x-y) \eta(y)$$

Integration order doesn't matter so

$$it's: \int_x \int_y \bar{\eta}(x) S_F(x-y) \eta(y)$$

$$\Rightarrow i \int_x \bar{\eta} \psi' + \bar{\psi}' \eta = -i \int_x \bar{\eta} \psi + \bar{\psi} \eta + 2i \int_x \int_y \bar{\eta}(x) i S_F(x-y) \eta(y)$$

Now we calculate $\int_x \int_y \bar{\psi}'(x) i S_F^{-1}(x-y) \psi'(y)$

$$= \int_x \int_y - \left[\bar{\psi}(x) - i \int_{z'} \bar{\eta}(z') S_F(z'-x) \right] i S_F^{-1}(x-y) \left[\psi(y) - i \int_{z'} S_F(y-z') \eta(z') \right]$$

$$= + \int_x \int_y \bar{\psi}(x) i S_F^{-1}(x-y) \psi(y) - i \int_x \int_y \int_{z'} \bar{\eta}(z') S_F(z'-x) i S_F^{-1}(x-y) \psi(y) \quad \leftarrow \text{We first integrate this over } x \text{ to get delta function.}$$

$$- i \int_x \int_y \int_{z'} \bar{\psi}(x) i S_F^{-1}(x-y) S_F(y-z') \eta(z') \quad \uparrow \text{First integrate over } y$$

$\uparrow$ integrate over y

$$\Rightarrow = \int_x \int_y \bar{\Psi}(x) i S_F^{-1}(x-y) \Psi(y) - i^2 \int_y \int_z \bar{\eta}(z) \delta(z-y) \Psi(y) - i^2 \int_x \int_{z'} \bar{\Psi}(x) \delta(x-z') \eta(z') \\ + i^2 \int_x \int_z \int_{z'} \bar{\eta}(z) S_F(z-x) \delta(x-z') \eta(z')$$

Now we integrate over the delta-functions

$$\Rightarrow = \int_x \int_y \bar{\Psi}(x) i S_F^{-1}(x-y) \Psi(y) + \int_y \bar{\eta}(y) \Psi(y) + \int_x \bar{\Psi}(x) \eta(x) - i \int_x \int_z \bar{\eta}(z) S_F(z-x) \eta(x)$$

Let's rename the variables that are integrated over [won't change anything]

$$= \int_x \int_y \bar{\Psi}(x) i S_F^{-1}(x-y) \Psi(y) + \int_x [\bar{\eta}(x) \Psi(x) + \bar{\Psi}(x) \eta(x)] - i \int_x \int_y \bar{\eta}(x) S_F(x-y) \eta(y)$$

Now we put the result to the generating functional:

$$Z_0[\bar{\eta}, \eta] = \int D\Psi D\bar{\Psi} \exp \left[i \int_x \int_y \bar{\Psi}(x) i S_F^{-1}(x-y) \Psi(y) + i \int_x [\bar{\eta} \Psi + \bar{\Psi} \eta] \right]$$

$$Z_0[\bar{\eta}, \eta] = \int D\Psi D\bar{\Psi} \exp \left[i \left(\int_x \int_y \bar{\Psi}(x) i S_F^{-1}(x-y) \Psi(y) + \int_x [\bar{\eta} \Psi + \bar{\Psi} \eta] - i \int_x \int_y \bar{\eta}(x) S_F(x-y) \eta(y) \right) \right. \\ \left. + i \left(- \int_x [\bar{\eta} \Psi + \bar{\Psi} \eta] + 2i \int_x \int_y \bar{\eta}(x) S_F(x-y) \eta(y) \right) \right]$$

We sum terms inside the exponentials and get

$$Z_0[\bar{\eta}, \eta] = \underbrace{\int D\Psi D\bar{\Psi} \exp \left[i \int_x \int_y \bar{\Psi}(x) i S_F^{-1}(x-y) \Psi(y) \right]}_{\text{this term is } Z_0[0,0]} \exp \left[i \int_x \int_y \bar{\eta}(x) i S_F(x-y) \eta(y) \right]$$

$$\Rightarrow Z_0[\bar{\eta}, \eta] = Z_0[0,0] \exp \left[- \int d^4x d^4y \bar{\eta}(x) S_F(x-y) \eta(y) \right] \quad \square$$

Exercise 2.7 The Yukawa theory is defined by the Lagrangian

$$\mathcal{L} = \mathcal{L}_{\text{kin}} + \mathcal{L}_\phi + g \phi \bar{\Psi} \Psi$$

The generating functional is $Z[J, \eta, \bar{\eta}] = \exp \left[-g \int_z \frac{\delta}{\delta J_z} \frac{\delta}{\delta \eta_z} \frac{\delta}{\delta \bar{\eta}_z} \right] Z_{\text{kin}}[J] Z_0[\eta, \bar{\eta}]$

$$Z_{\text{kin}}[J] = \exp \left[\frac{i}{2} \int_x \int_y J(x) i \Delta_F(x-y) J(y) \right] \quad \text{and} \quad Z_0[\eta, \bar{\eta}] = \exp \left[i \int_x \int_y \bar{\eta}(x) i S_F(x-y) \eta(y) \right]$$

For $Z[0,0,0]$ all odd derivatives vanish (since we will have $J=0=\eta=\bar{\eta}$)

We expand the exponent and the first non-zero contribution comes from the second order term. In the second order we get

$$Z[J, \eta, \bar{\eta}] = \frac{(-g)^2}{2} \int_x \int_y \frac{\delta}{\delta J(x)} \frac{\delta}{\delta J(y)} Z_{KG}[J] \frac{\delta}{\delta \eta_a(x)} \frac{\delta}{\delta \bar{\eta}_a(x)} \frac{\delta}{\delta \eta_b(y)} \frac{\delta}{\delta \bar{\eta}_b(y)} Z_D[\eta, \bar{\eta}] + \mathcal{O}(g^4)$$

Here the indices a and b are for the different spinors and repeated indices are summed over

$$\Rightarrow \Psi_a \bar{\Psi}_a = \sum_{a=1}^4 \Psi_a \bar{\Psi}_a$$

Now to calculate $Z[0,0,0]$ we compute the functional derivative and then put the sources to be zero at the end.

Let's first calculate the Klein-Gordon part

$$\frac{\delta}{\delta J(x)} Z_{KG}[J] = \frac{\delta}{\delta J(x)} e^{\frac{i}{2} \int_k \int_l J(k) i \Delta_F(k-l) J(l)}$$

We use the axioms:

$$\frac{\delta J(y)}{\delta J(x)} = \delta^{(4)}(x-y) \quad \text{and}$$

$$= \frac{i}{2} \left[\int_z i \Delta_F(x-z) J(z) + J(z) i \Delta_F(z-x) \right] Z_{KG}[J] \quad \left| \frac{\delta}{\delta J(x)} \int d^4y J(y) \phi(y) = \phi(x) \right.$$

$$\Rightarrow \frac{\delta}{\delta J(x)} Z_{KG}[J] = \frac{i^2}{2} \int_z \left[\Delta_F(x-z) J(z) + J(z) \Delta_F(z-x) \right] Z_{KG}[J]$$

$$\Rightarrow \frac{\delta}{\delta J(x)} \frac{\delta}{\delta J(y)} Z_{KG}[J] = \frac{\delta}{\delta J(x)} \left[\frac{i^2}{2} \int_z (\Delta_F(y-z) J(z) + J(z) \Delta_F(z-y)) Z_{KG}[J] \right]$$

$$= \frac{i^2}{2} (\Delta_F(y-x) + \Delta_F(x-y)) Z_{KG}[J] + \text{Term that goes to 0 when } J=0$$

$$\Rightarrow \frac{\delta}{\delta J(x)} \frac{\delta}{\delta J(y)} Z_{KG}[J] \Big|_{J=0} = \frac{i^2}{2} (\Delta_F(y-x) + \Delta_F(x-y)) = i^2 \Delta_F(x-y) \quad \parallel \Delta_F(x-y) = \Delta_F(y-x)$$

Now we calculate the Dirac part, we have additional functional derivative rule of

$$\frac{\delta \eta_b(y)}{\delta \bar{\eta}_a(x)} = \delta(x-y) \delta_{ab}$$

$$= S_F^{cd}(u-v)$$

$$\Rightarrow \frac{\delta}{\delta \bar{\eta}_b(y)} Z_D[\eta, \bar{\eta}] = \frac{\delta}{\delta \bar{\eta}_b(y)} e^{i \int_u \int_v \bar{\eta}(u) i S_F(u-v) \eta(v)} = \left[\frac{\delta}{\delta \bar{\eta}_b(y)} i \int_u \int_v \bar{\eta}_c(u) i S_F^{cd}(u-v) \eta_d(v) \right] Z_D[\eta, \bar{\eta}]$$

$$\frac{\delta}{\delta \bar{\eta}_b(y)} Z_D[\eta, \bar{\eta}] = \left[i^2 \int_v \delta_{cb} S_F^{cd}(y-v) \eta_d(v) \right] Z_D[\eta, \bar{\eta}] = \left(i^2 \int_v S_F^{bd}(y-v) \eta_d(v) \right) Z_D[\eta, \bar{\eta}]$$

$$\Rightarrow \frac{\delta}{\delta \eta_b(y)} \frac{\delta}{\delta \bar{\eta}_b(y)} Z_D[\eta, \bar{\eta}] = i^2 S_F^{bb}(y-y) Z_D[\eta, \bar{\eta}] + i^2 \int S_F^{bd}(y-v) \eta_d(v) \int_u \bar{\eta}_c(u) S_F^{cb}(u-y) Z_D[\eta, \bar{\eta}]$$

$$\frac{\delta}{\delta \bar{\eta}_a(x)} \frac{\delta}{\delta \eta_b(y)} \frac{\delta}{\delta \bar{\eta}_b(y)} Z_D[\eta, \bar{\eta}] = i^2 i^2 \left[\int_F^{bb}(0) \int_V S_F^{ad}(x-v) \eta(v) \right] Z_D[\eta, \bar{\eta}] + i^2 i^2 \left[\int_V S_F^{bd}(y-v) \eta(v) \right] S_F^{ab}(x-y) Z_D[\eta, \bar{\eta}]$$

+ Higher order term that goes to 0 when $\eta = \bar{\eta} = 0$

$$\Rightarrow \frac{\delta}{\delta \eta_a(x)} \frac{\delta}{\delta \bar{\eta}_a(x)} \frac{\delta}{\delta \eta_b(y)} \frac{\delta}{\delta \bar{\eta}_b(y)} Z_D[\eta, \bar{\eta}] = \int_F^{bb}(0) S_F^{aa}(0) Z_D[\eta, \bar{\eta}] + \int_F^{ba}(y-x) S_F^{ab}(x-y) Z_D[\eta, \bar{\eta}] + 0$$

$$\Rightarrow Z[0,0,0] = \left(-\frac{g^2}{2} \int_x \int_y \left[\frac{\delta}{\delta S(x)} \frac{\delta}{\delta S(y)} Z_{K_0}[J] \frac{\delta \delta \delta \delta}{\delta \eta_a(x) \delta \bar{\eta}_a(x) \delta \eta_b(y) \delta \bar{\eta}_b(y)} Z_D[\eta, \bar{\eta}] \right] \right) \Big|_{J=\eta=\bar{\eta}=0}$$

Now we substitute the results of derivatives

$$Z[0,0,0] = \frac{g^2}{2} \int_x \int_y i^2 \Delta_F(x-y) \underbrace{\int_F^{bb}(0)}_{\text{Trace}} \underbrace{S_F^{aa}(0)}_{\text{Trace}} Z_D[0,0] + \underbrace{\int_F^{ba}(y-x) S_F^{ab}(x-y)}_{\text{Trace}} Z_D[0,0] \Bigg| Z_D[0,0] =$$

$$Z[0,0,0] = -\frac{g^2}{2} \int_x \int_y \Delta_F(x-y) \left[\text{Tr}(S_F(y-x) S_F(x-y)) + \text{Tr}(S_F(0)) \text{Tr}(S_F(0)) \right]$$

The normalization factor $Z[0,0,0]$ to $\mathcal{O}(g^2)$ is

$$Z[0,0,0] = -\frac{g^2}{2} \int_x \int_y \Delta_F(x-y) \left[\text{Tr}(S_F(y-x) S_F(x-y)) + \text{Tr}(S_F(0)) \text{Tr}(S_F(0)) \right]$$